2. Read the following passage carefully : *10*

(1) Rotation of crops is a universal phenomenon which is practised by most of the farmers of the tropical and temperate countries. The main objective of rotation of crops is to obtain higher agricultural returns on the one hand, and to maintain the soil fertility on the other.

(2) In other words, the rotation of crops helps in making agriculture more sustainable. The importance of crop rotation is more in the areas where farmers grow two, or more than two crops in the same field in a year. Irrigation facilities have also been appreciably developed in the country during the last three decades. The availability of water to the arable land has helped in the intensification of agriculture.

(3) In the areas such as Punjab and Haryana, where the Green Revolution is a big success, one soil exhaustive crop (rice) is followed by another soil exhaustive crop (wheat). Subsequently, the field vacated by wheat is devoted to either rice or maize or cotton. Thus, in one year, the farmers are harvesting three soil exhaustive crops from the same field. Such a rotation of crops may fetch more income to the farmers, but depletes the soil fertility at a faster pace.

(4) A number of field studies were conducted to assess the traditional crop rotation pattern. One such field study about the changes in the rotation of crops was conducted in the village Banhera (Tanda), Haridwar district. The main rotation of crops of the village is shown in the table below.

Traditional Rotation of Crops (1960-65) in Banhera (Tanda)				
Year	*Kharif (mid-June to mid-October)*	*Rabi (mid-October to mid-April)*	*Zaid (April to June)*	*No. of days land left fallow*
1960	Millet/fodder/rice	Gram	Fallow	90
1961	Fallow	Wheat	Fallow	210
1962	Millet/fodder/rice	Gram	Fallow	90
1963	Fallow	Wheat	Fallow	210
1964	Millet mixed with urad/fodder/rice	Gram	Fallow	90
1965	Fallow	Wheat	Fallow	210

Answer the following questions, based on the above passage :

(i) Fill in the blanks with the appropriate option from those given in brackets, based on your understanding of para 2. *1*

One of the reasons that there has been an intensification of agriculture in the country is the development of irrigation facilities is a/an _____ (fact/opinion) because it is a/an _____ (subjective judgement/objective detail).

(ii) Based on your understanding of paras 1 and 2, state whether the following statement is true or false. *1*

If proper irrigation facility is provided, higher agricultural returns can be obtained by making crop rotation more sustainable.

(iii) Justify the following in about 40 words.

Crop rotation may fetch more income to the farmers, but depletes the soil fertility at a faster pace. *2*

(iv) Based on the table, mention the years when the land was left fallow for maximum number of days. *1*

(v) The study conducted in village Banhera (Tanda) reflects that during the years 1960 to 1965, from April to June, the land was left fallow or uncultivated throughout. State any one inference that can be drawn from this. *1*

(vi) Select the option that correctly states the significance of crop rotation as given below : *1*

(1) Crop rotation is done to obtain higher agricultural returns.

(2) Higher agricultural returns can be obtained by harvesting soil-exhaustive crops.

(3) Crop rotation demands that land should be left fallow for a few months to increase the fertility of soil.

(4) Better irrigation facility to the arable land has helped in the intensification of agriculture.

(5) If crops to be harvested are chosen wisely, crop rotation can help in maintaining fertility of soil.

Options :

(A) (1), (2) and (4)

(B) (1) only

(C) (3) and (5)

(D) (1) and (5)

(vii) Crop rotation is helping in making agriculture more sustainable. Mention one benefit and one drawback of crop rotation. (Answer in about 40 words) *2*

(viii) Which of the following is the main takeaway from the study mentioned in the passage ? *1*

(A) The study needs to include more valid data to support the practice of crop rotation.

(B) Crop rotation is a sustainable practice that can transform the agricultural sector.

(C) The impact of crop rotation on farmer's upliftment has been thoroughly explained.

(D) Crop rotation is independent of the type of crops harvested and availability of water for irrigation.